Logarithmic equations

The domain first, the laws second, the check last — and no step in a different order.

LESSON 61–62 | 2 × 40 MIN | ALGEBRA 10, §3.6 | P2 · 2.4

LESSON CLOCK

12' | 22' | 22' | 22' | 20' | 12'

The domain first	12 min
Equating arguments	22 min
Exponential form	22 min
Substitution	22 min
Practice	20 min
Homework	12 min

LEARNING OBJECTIVES

- State the domain of a logarithmic equation before solving it.
- Solve log_a f(x) = log_a g(x) by equating arguments.
- Solve equations by converting to exponential form.
- Reject roots that fall outside the domain.

TEXTBOOK

National	pp. 249–260
Cambridge	Pure Mathematics 2 & 3, pp. 39–44
Workbook	P2 Exercise 2F

01 Explanation

The domain first

EVERY ARGUMENT MUST BE POSITIVE

Before any algebra, write the conditions. For $log_2(x - 1) + log_2(x + 2) = 2$: $x - 1 > 0$ and $x + 2 > 0$, so $x > 1$.

Condensing the two logarithms into $log_2[(x-1)(x+2)]$ **widens** the domain — the product is also positive when both factors are negative. That is exactly where extraneous roots come from, and the domain line written first is what removes them.

THE CONDENSING STEP IS NOT REVERSIBLE

$log A + log B = log(AB)$ is true only where both A and B are positive. Used left to right in an equation, it is a one-way street.

Equating arguments

> $log\_a\ f(x) = log\_a\ g(x) \Rightarrow f(x) = g(x)$

Valid because the logarithm is one-to-one — the same property that solved exponential equations, used in the other direction.

$$\log_3(2x - 1) = \log_3(x + 4) \Rightarrow 2x - 1 = x + 4 \Rightarrow x = 5$$

Check: $2(5) - 1 = 9 > 0$ and $5 + 4 = 9 > 0$ ✓.

Converting to exponential form

When one side is a number rather than a logarithm, use the definition:

> $log\_a\ f(x) = c \Rightarrow f(x) = a\text{^}c$

$$\log_2(x - 1) + \log_2(x + 2) = 2 \Rightarrow \log_2[(x-1)(x+2)] = 2 \Rightarrow (x-1)(x+2) = 4$$

So $x^2 + x - 6 = 0$, giving $x = 2$ or $x = -3$. The domain was $x > 1$, so $x = -3$ is rejected and only $x = 2$ survives.

THE SHAPE OF A FULL SOLUTION

1 Domain. **2** Condense to a single logarithm. **3** Convert or equate. **4** Solve. **5** Reject anything outside the domain, and say why.

Substitution

When the same logarithm appears twice, name it:

> $(\lg x)^2 - 3 \lg x + 2 = 0 \text{ , } t = \lg x$

Then $t^2 - 3t + 2 = 0$ gives $t = 1$ or $t = 2$, so $x = 10$ or $x = 100$. Both are positive, so both stand.

Note that t here has **no** sign restriction — a logarithm may be any real number. That is the opposite of the exponential substitution, and confusing the two is a common error.

02 Worked examples

EXAMPLE 1 **Solve $\log_5(3x - 2) = 2$.**

1. Domain: $3x - 2 > 0 \Rightarrow x > \frac{2}{3}$.
2. $3x - 2 = 25$
3. $x = 9$. In the domain ✓

ANSWER

$x = 9$

EXAMPLE 2 **Solve $\log_2(x - 1) + \log_2(x + 2) = 2$.**

1. Domain: $x > 1$.
   Both arguments positive.
2. $(x - 1)(x + 2) = 4$
3. $x^2 + x - 6 = 0 \Rightarrow x = 2, -3$
4. $x = -3$ fails the domain.

ANSWER

$x = 2$

EXAMPLE 3 **Solve $(\lg x)^2 - 3\lg x + 2 = 0$.**

1. Domain $x > 0$. Let $t = \lg x$.
   No sign condition on t.
2. $t^2 - 3t + 2 = 0$
3. $t = 1, 2$
4. $x = 10$ or $x = 100$.

ANSWER

$x = 10$ and $x = 100$

03 Interactive model

Write the domain in a box at the top of the board and leave it there.

Domain first

Domain of $log_2(x - 3) = 1$:

- $x > 0$
- $x > 3$
- $x \geq 3$
- $\mathbb{R}$

$log_5(3x - 2) = 2$ gives:

- $x = 9$
- $x = 4$
- $x = 27$
- $x = 3$

$log_2(x-1) + log_2(x+2) = 2$ gives:

- $x = 2, -3$
- $x = 2$
- $x = -3$
- no solution

In $t = lg\ x$ the condition on t is:

- $t > 0$
- $t \geq 0$
- none
- $t \neq 0$

$log_3(2x-1) = log_3(x+4)$ gives:

- $x = 5$
- $x = 3$
- $x = 1$
- no solution

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Logarithmic equation	Logarifmik tenglama	Логарифмическое уравнение
Domain of definition	Aniqlanish sohasi	Область допустимых значений
Equating arguments	Argumentlarni tenglashtirish	Приравнивание аргументов
Exponential form	Ko‘rsatkichli shakl	Показательная форма
Extraneous root	Chet ildiz	Посторонний корень
Condensing	Bir logarifmga keltirish	Свёртывание
Substitution	Almashtirish	Замена
Verification	Tekshirish	Проверка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The first line of a logarithmic equation is:

- the law used
- the domain
- the answer
- the check

$log\_a\ f = log\_a\ g$ gives $f = g$ because the logarithm is:

- positive
- one-to-one
- continuous
- even

Condensing two logarithms:

- never changes the domain
- can widen the domain
- narrows the domain
- is illegal

$\log_2 x = 5$ gives:

- $x = 10$
- $x = 25$
- $x = 32$
- $x = 2.5$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Solve $log_2 x = 4$

ANSWER $x = 16$

2. Solve $log_3 x = 2$

ANSWER $x = 9$

3. Solve $lg\, x = 3$

ANSWER $x = 1000$

4. Solve $log_5(x + 1) = 1$

ANSWER $x = 4$

5. Domain of $log_2(x - 5)$

ANSWER $x > 5$

6. Solve $log_2 x = 0$

ANSWER $x = 1$

7. Solve $log_4 x = \frac{1}{2}$

ANSWER $x = 2$

Medium · the standard question

7 PROBLEMS

1. Solve $\log_5(3x - 2) = 2$

ANSWER $x = 9$

2. Solve $\log_3(2x - 1) = \log_3(x + 4)$

ANSWER $x = 5$

3. Solve $\log_2(x - 1) + \log_2(x + 2) = 2$

ANSWER $x = 2$

4. Solve $\lg x + \lg(x - 3) = 1$

ANSWER $x = 5$

5. Solve $(\lg x)^2 - 3\lg x + 2 = 0$

ANSWER $x = 10, 100$

6. Solve $\log_2(x + 3) - \log_2(x - 1) = 2$

ANSWER $x = \frac{7}{3}$

7. Solve $\log_3(x^2) = 4$

ANSWER $x = \pm 9$

Hard · stretch and reason

7 PROBLEMS

1. Solve $\log_2(x - 2) + \log_2(x - 3) = 1$

   ANSWER $x = 4$

2. Solve $(\log_2 x)^2 - \log_2 x - 6 = 0$

   ANSWER $x = 8$ or $x = \frac{1}{4}$

3. Solve $\lg(x + 6) - \frac{1}{2} \lg(2x - 3) = 2 - \lg 25$

   ANSWER $x = 6$

4. Solve $\log_x 4 + \log_x 16 = 3$

   ANSWER $x = 4$

5. Solve $\log_3 x + \log_x 3 = 2.5$

   ANSWER $x = 9$ or $x = \sqrt{3}$

6. Solve $\lg^2 x - \lg x^3 + 2 = 0$

   ANSWER $x = 10, 100$

7. Explain why $\lg x + \lg(x - 3) = 1$ has only one root

   ANSWER $x = -2$ fails the domain

07 Homework

Homework — 6 tasks

Open every solution with the domain, and close it with the rejection.

1. Solve $\log_4(2x + 3) = 2$.
2. Solve $\log_2(x + 5) = \log_2(3x - 1)$.
3. Solve $\lg x + \lg(x - 15) = 2$.
4. Solve $(\log_3 x)^2 - 4 \log_3 x + 3 = 0$.

5. Solve $log_5(x + 4) - log_5(x - 2) = 1$.
6. Explain in three sentences why condensing two logarithms can create a root the original never had.